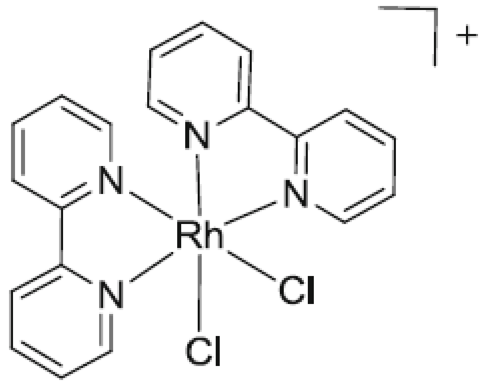
Organometallic Complexes

Ligand	Ionic model		Covalent Model
	Charge	Donated e^-	Donated e^-
H (terminal hydride)	−1	2	1
μ-H (bridging halide)	−1	2/2[a]	1/2[a]
X (terminal F, Cl, Br, I, CN, etc.)	−1	2	1
μ-X (bridging halide)	−1	2/2[a]	1/2[a]
ER_2 (E = O, S, Se, Te)	0	2	2
OR, SR (terminal alkoxide or thiolate)	−1	2	1
μ-OR, μ-SR (bridging alkoxide or thiolate)	−1	2/2[a]	1/2[a]
NR_3, PR_3 (terminal amine or phosphine)	0	2	2
NR_2, PR_2 (terminal amide or phosphide)	−1	2	1
μ-NR_2, μ-PR_2 (bridging amide or phosphide)	−1	2/2[a]	1/2[a]
NCR (nitrile)	0	2	2
CH_3 or other alkyl	−1	2	1
μ-CH_3 or other alkyl (bridging alkyl)	−1	2/2[a]	1/2[a]
η^1-aryl, alkenyl, alkynyl	−1	2	1
η^3-allyl	−1	4	3
η^4-diene	0	4	4
η^5-cyclopentadienyl (η^5-Cp)	−1	6	5
η^6-arene	0	6	6
η^1-acyl	−1	2	1
η^2-ketone	0	2	2
η^2-alkene	0	2	2
η^2-alkyne	0	2	2
μ-η^2-alkyne	0	2/2[a]	2/2[a]
CO (carbonyl)	0	2	2
μ-CO (bridging carbonyl)	0	1/1[a]	1/1[a]
CNR (isocyanide, isonitrile)	0	2	2
NO (linear nitrosyl)	+1	2	3
NO (bent nitrosyl)	−1	2	1
CR_2 (terminal alkylidene or Schrock carbene)	−2	4	2
CR (terminal akylidyne or Schrock carbyne)	−3	6	3
CR_2 (terminal Fisher carbene)	0	2	2
CR (terminal Fisher carbyne)	−1	4	3

Example 1: $[RhCl_2(bpy)_2]^+$



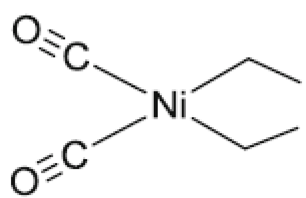
Ionic model:

Rh(III)	d^6	6 e^-
bpy	4 e^- x 2	8 e^-
Cl^-	2 e^- x 2	4 e^-
		18 e^-

Covalent model:

Rh(0)	d^9	9 e^-
bpy	4 e^- x 2	8 e^-
Cl·	1 e^- x 2	2 e^-
		19 e^-
(overall +1 charge)		– 1 e^-
		18 e^-

Example 2: $[Ni(Et)_2(CO)_2]$



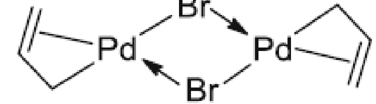
Ionic model:

Ni(II)	d^8	8 e^-
CO	2 e^- x 2	4 e^-
Et^-	2 e^- x 2	4 e^-
		16 e^-

Covalent model:

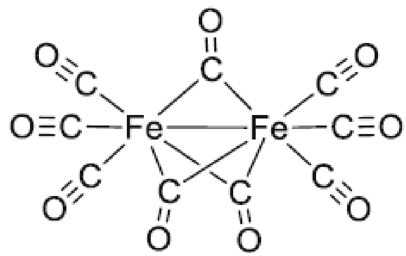
Ni(0)	d^{10}	10 e^-
CO	2 e^- x 2	4 e^-
Et·	1 e^- x 2	2 e^-
		16 e^-

Example 1: $[Pd(\eta^3\text{-allyl})Br]_2$



Ionic model:			***Covalent model:***		
Pd(II)	d^8	8 e^-	Pd(0)	d^{10}	10 e^-
allyl$^-$	4 e^- x 1	4 e^-	allyl·	3 e^- x 1	3 e^-
Br^-	2 e^- x 1	2 e^-	Br·	1 e^- x 1	1 e^-
μ-Br	2 e^- x 1	2 e^-	μ-Br	2 e^- x 1	2 e^-
		16 e^-			**16 e^-**

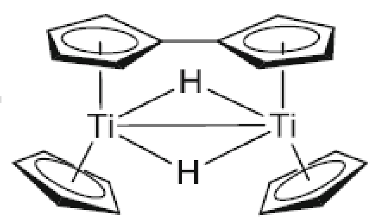
Example 2: $[Fe_2(CO)_9]$



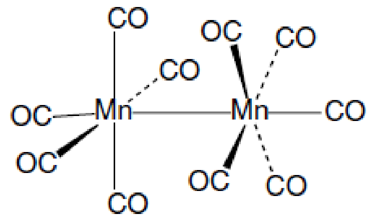
Ionic & Covalent model:

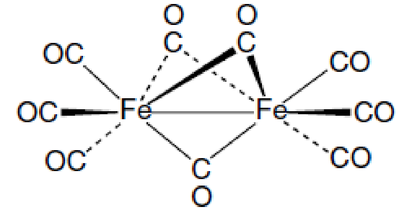
Fe(0)	d^8	8 e^-
CO	2 e^- x 3	6 e^-
μ-CO	1 e^- x 3	3 e^-
M–M	1 e^-	1 e^-
		18 e^-

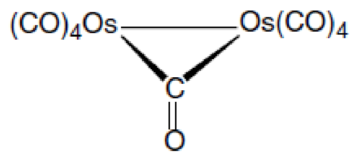
Example 3: $[Ti(\eta^5\text{-Cp})H]_2(Cp\text{-}Cp)$

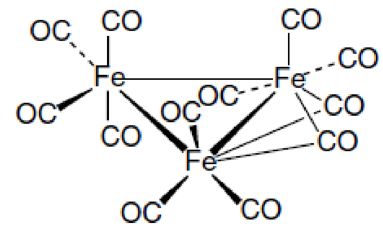


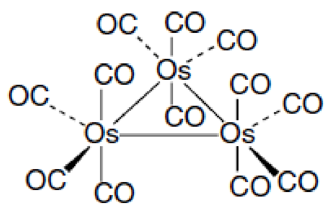
Ionic model:			***Covalent model:***		
Ti(III)	d^1	1 e^-	Ti(0)	d^4	4 e^-
Cp^-	6 e^- x 2	12 e^-	Cp·	5 e^- x 2	10 e^-
H^-	2 e^- x 1	2 e^-	H·	1 e^- x 1	1 e^-
μ-H	2 e^- x 1	2 e^-	μ-H	2 e^- x 1	2 e^-
M–M	1 e^-	1 e^-	M–M	1 e^-	1 e^-
		18 e^-			**18 e^-**

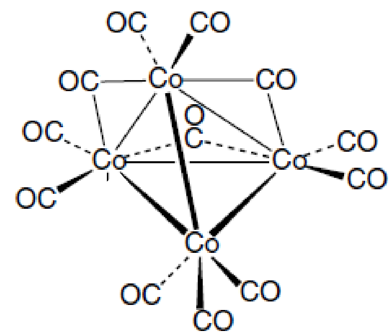


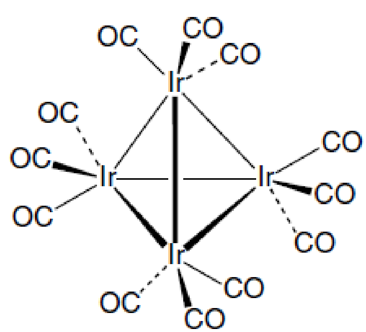


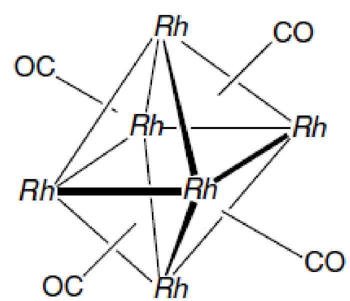








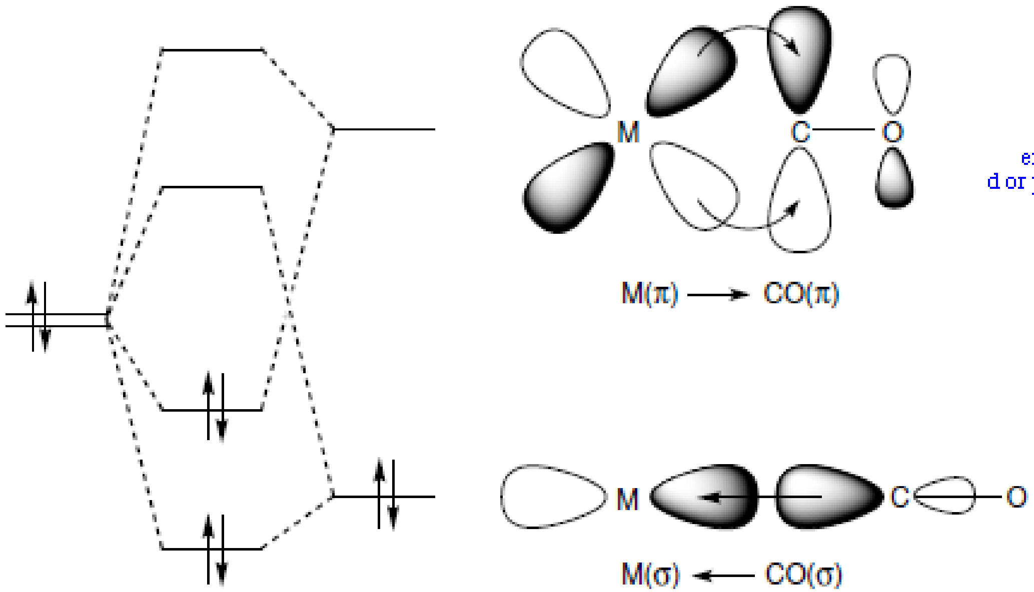


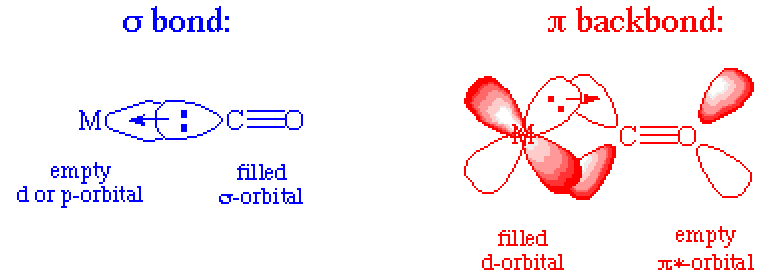


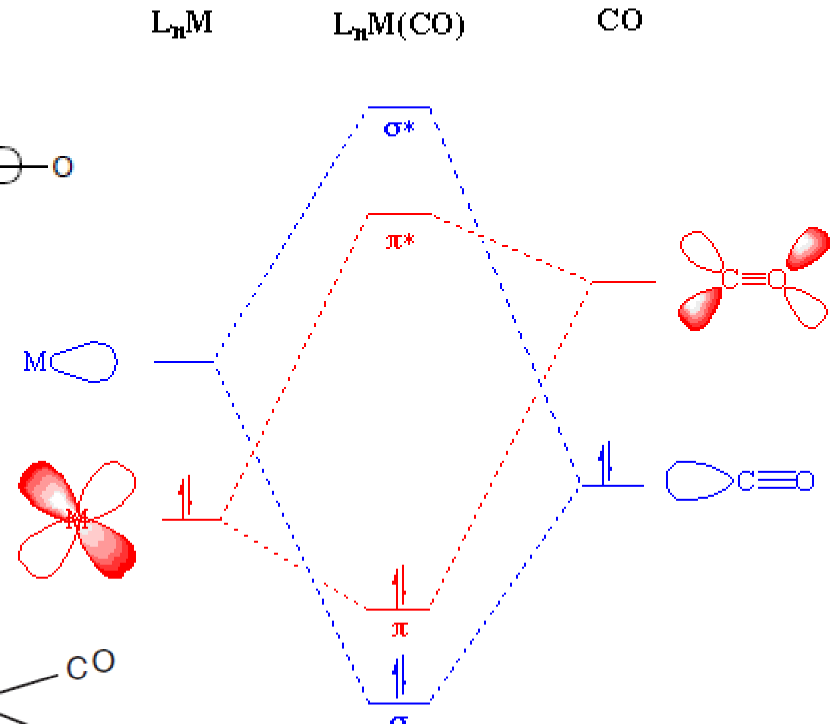
Rh = $Rh(CO)_2$

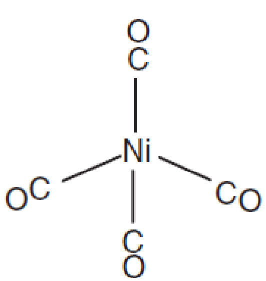
$Rh_6(CO)_{16} = Rh_6(CO)_{12}(\mu_3\text{-}CO)_4$

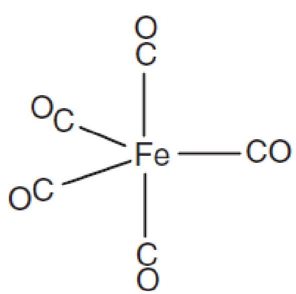
CO-Bonding Modes

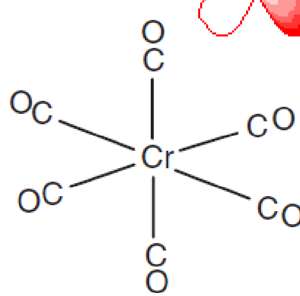


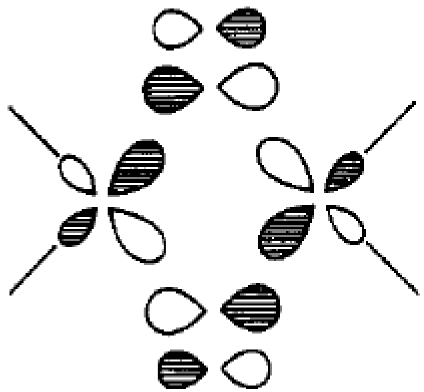


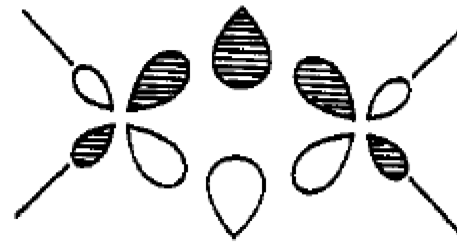


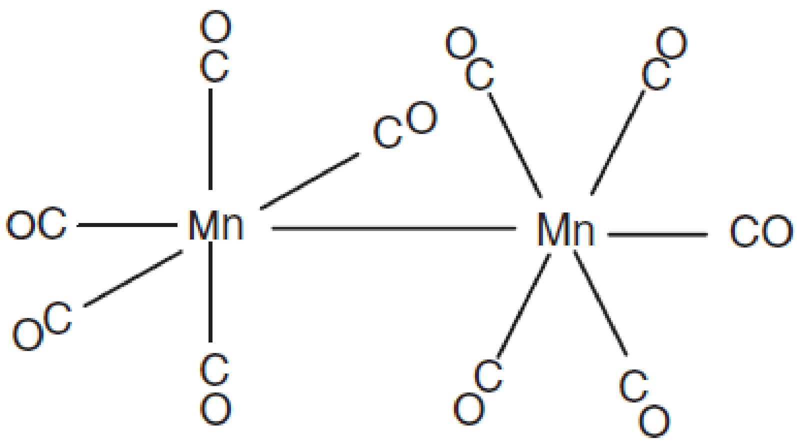


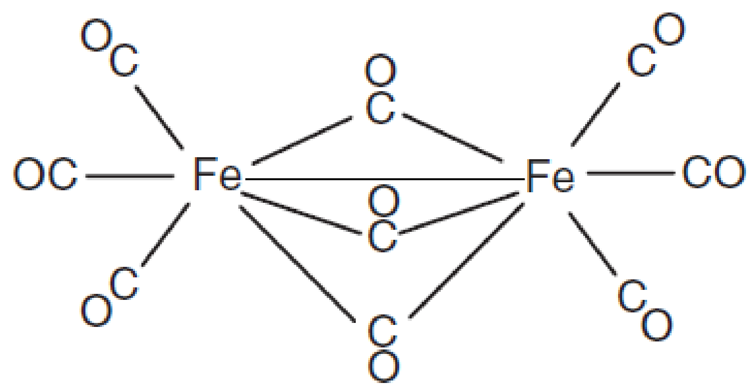


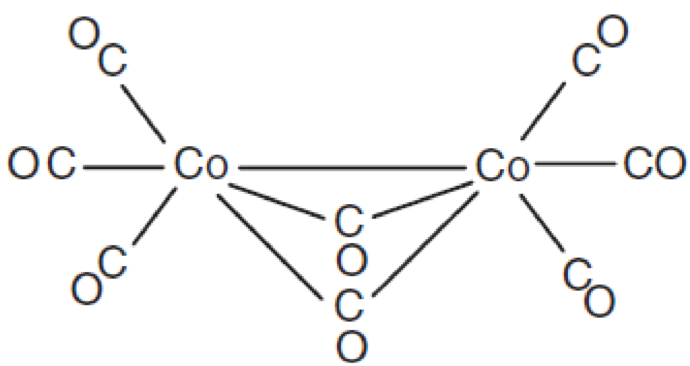


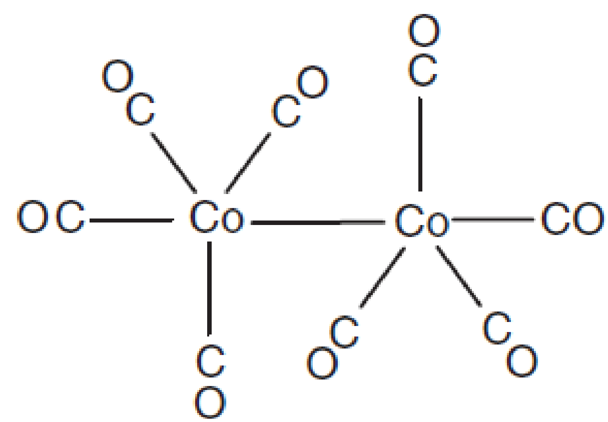


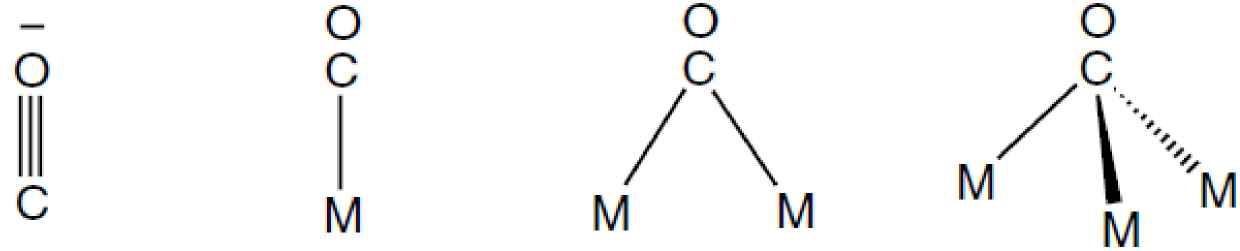












	free	terminal	μ_2-CO	μ_3-CO
ν_{CO} (cm^{-1})	2143	1850-2120	1750-1850	1620-1730